

Satellite Image Land-Use Classifier & Temporal Change Detector

Project Report

1. Introduction & Objective

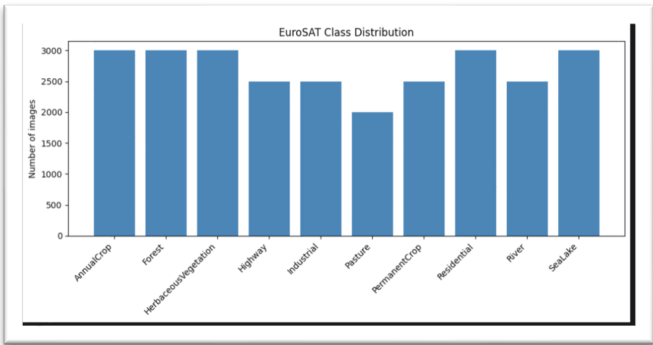
This project builds a computer vision pipeline that (a) classifies satellite tiles into land-use categories using transfer learning, and (b) detects land-cover change between two time periods using embedding similarity, wrapped in an interactive dashboard. The pipeline was built and trained in Google Colab across five notebooks, covering data preparation, a scratch baseline model, fine-tuned transfer learning, temporal change simulation, and a final evaluation stage.

2. Data Pipeline

EuroSAT (27,000 RGB tiles, 10 land-use classes) was loaded via torchvision and split using an approximated spatial block split: since the standard release has no geographic coordinates, each class's images were kept in their original order and divided into contiguous chunks (70% / 15% / 15%) instead of a random shuffle, to reduce spatial leakage between train, validation, and test.

Split	Size
Train	18,900
Validation	4,050
Test	4,050

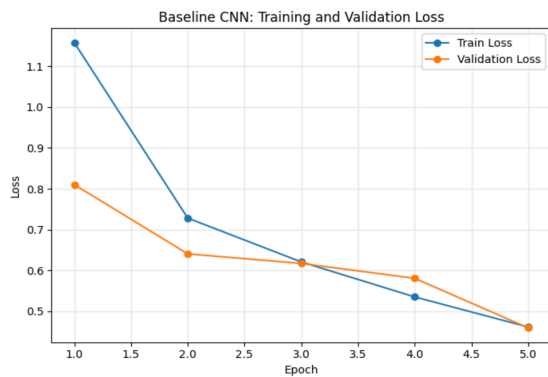
Figure 1: Class Distribution



3. Baseline CNN

A simple 3-layer convolutional network was trained from scratch as the performance floor for comparison against transfer learning.

Figure 2: Baseline Training Curve



Per-Class F1 (Validation)

Class	F1 Score
AnnualCrop	0.857
Forest	0.923
HerbaceousVegetation	0.790
Highway	0.645
Industrial	0.929
Pasture	0.794
PermanentCrop	0.739
Residential	0.946
River	0.708
SeaLake	0.930

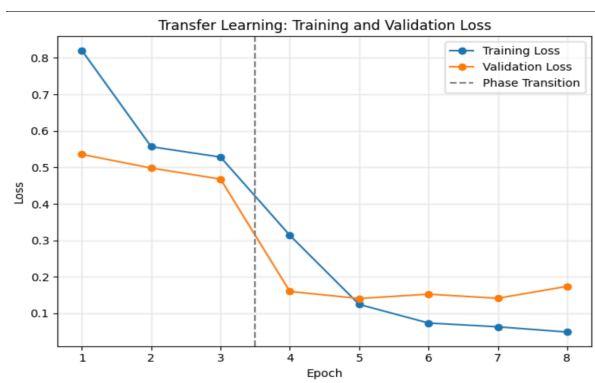
Macro F1: 0.826

Highway (0.645) and River (0.708) were the weakest classes for the scratch baseline, likely due to their visual overlap with Industrial/urban tiles and other water/vegetation classes respectively at low resolution.

4. Transfer Learning (ResNet18)

ResNet18 was fine-tuned using a two-phase strategy: Phase 1 froze the backbone and trained only the classifier head (3 epochs); Phase 2 unfroze the last two residual blocks and continued training with a 10x lower learning rate (5 more epochs).

Figure 3: Fine-Tuning Training Curve



Frozen vs. Unfrozen Ablation

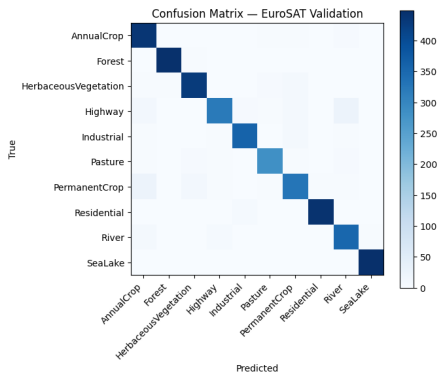
Setting	Macro F1
Frozen backbone (Phase 1)	0.849
Unfrozen last 2 blocks (Phase 2)	0.948

Unfreezing the last two residual blocks improved macro F1 by roughly 10 points (0.849 $\rightarrow$ 0.948) over the frozen-head-only baseline, and both settings substantially outperform the scratch CNN (0.826), confirming that ImageNet-pretrained features transfer well to satellite imagery, and that allowing some backbone adaptation helps further.

Per-Class F1 (EuroSAT Validation, Phase 2)

Class	F1 Score
AnnualCrop	0.930
Forest	0.990
HerbaceousVegetation	0.955
Highway	0.901
Industrial	0.963
Pasture	0.950
PermanentCrop	0.895
Residential	0.984
River	0.922
SeaLake	0.993

Figure 4: EuroSAT Confusion Matrix



UC Merced Cross-Dataset Evaluation

The fine-tuned model (trained only on EuroSAT's 10 classes) was evaluated directly on all 2,100 images across UC Merced's 21 classes, without any label remapping — the model can only output one of its 10 known EuroSAT labels, so a prediction only counts as correct when it exactly matches the UC Merced class name (case-insensitive). This measures out-of-domain generalization rather than in-domain accuracy.

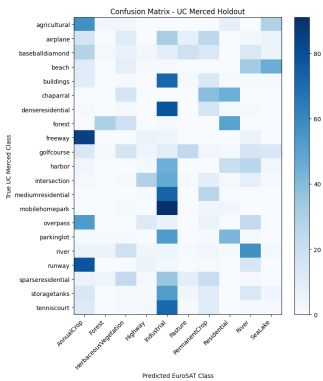
Metric	Value
Exact-name accuracy	0.041
Macro F1	0.037

Per-Class F1 (UC Merced, non-zero classes)

Class	F1 Score
forest	0.411
river	0.361
all other 19 classes	0.000

As expected, only the two UC Merced classes with a directly matching EuroSAT name (forest, river) score above zero — every other UC Merced class (e.g. airplane, golfcourse, tenniscourt) has no corresponding EuroSAT label, so the model can never predict it correctly under this exact-match rule. The near-zero overall macro F1 (0.037) is not a failure of the fine-tuned model itself; it reflects that EuroSAT and UC Merced use almost entirely different label vocabularies, and this evaluation method does not attempt semantic remapping between them.

Figure 5: UC Merced Confusion Matrix

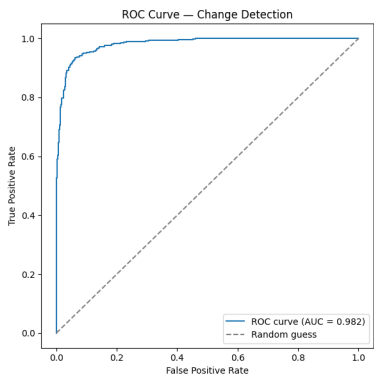


5. Temporal Change Detection

The fine-tuned ResNet18's classifier head was removed to produce a 512-dimensional embedding extractor, run over all 27,000 EuroSAT tiles. Since EuroSAT has no real before/after imagery, T1/T2 pairs were simulated: each class's images were split into contiguous T1 and T2 halves; unchanged pairs were drawn from the same class's T1/T2 pools, and changed pairs were drawn by pairing one class's T1 pool with the next class's T2 pool (cyclically). A pair's ground truth is 'changed' only if its two tiles come from different EuroSAT classes.

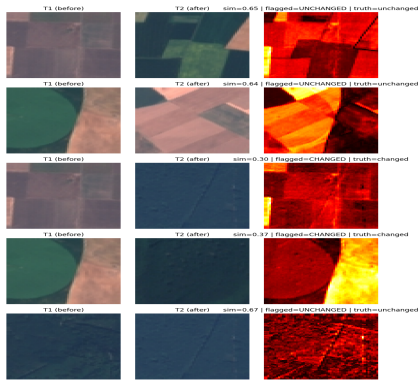
Metric	Value
Total simulated pairs	1,000
Unchanged pairs	500
Changed pairs	500
Cosine similarity range	0.150 – 0.975

Figure 6: ROC Curve



Using $1 - \text{cosine similarity}$ as the change score, Youden's J statistic (maximizing $\text{TPR} - \text{FPR}$) selected a change-score threshold of 0.536, equivalent to a similarity threshold of 0.464. At this operating point: $\text{TPR} = 0.936$, $\text{FPR} = 0.062$ — meaning the model correctly flags 93.6% of true changes while only false-alarming on 6.2% of unchanged pairs, a strong separation between the two classes under this simulation.

Figure 7: Sample Change Heatmaps



6. Spatial Leakage Experiment

To test whether the spatial block split actually reduces leakage compared to a random split, a fast baseline CNN was trained separately on (a) the block-split train/val and (b) a randomly shuffled train/val of the same ratio, and validation accuracy was compared.

Split Type	Validation Accuracy
Block split	0.801
Random split	0.767
Difference (random – block)	–0.034

This result runs opposite to the typical leakage hypothesis, which predicts random-split accuracy should look inflated (higher) relative to block-split accuracy if nearby, near-duplicate tiles are being split across train and validation. Here, block-split accuracy was actually higher by 3.4 points. Given the baseline CNN was only trained for 3 epochs per split, this gap is plausibly within normal run-to-run noise from random weight initialization and mini-batch ordering, rather than strong evidence against leakage. It may also reflect that the block split's particular validation chunk (a contiguous slice per class) happened to land on a comparatively 'easier' subset of images. A more conclusive test would repeat this comparison across multiple random seeds and report the average and spread, rather than a single run per split.

7. Error Analysis

The 5 most frequent (true class, predicted class) confusion pairs on the EuroSAT validation set (excluding correct predictions) were identified from the confusion matrix:

True Class	Predicted Class	Count
Highway	River	25
Permanent Crop	Annual Crop	25
Highway	Annual Crop	12
Permanent Crop	Herbaceous Vegetation	11
River	Annual Crop	9



Observations

The visual examples show that most classification errors occur between classes with similar visual characteristics from a satellite perspective. Possible reasons for confusion include:

- Similar colors and textures between different land-use categories.
- Limited spatial detail because images are resized to 64x64 pixels.
- Some classes share similar landscape patterns when viewed from above.

These confusion patterns suggest that additional training data, higher-resolution imagery, or more advanced architectures could further improve classification performance.

8. Dashboard

A Streamlit dashboard was built to make the trained model interactive: a user uploads a 'before' (T1) and 'after' (T2) tile and receives the predicted land-use class and confidence (plus top-3 predictions) for each tile, the cosine similarity between their embeddings, a change/no-change flag, and a pixel-difference heatmap. A sidebar toggle lets the user switch between three similarity-threshold operating points (High Recall, Balanced, High Precision) derived from the ROC analysis in Notebook 4.

Figure 9: Dashboard — Predictions

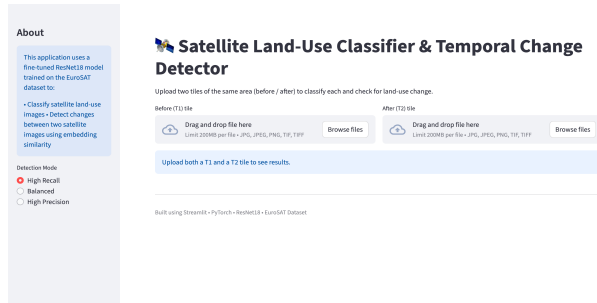


Figure 10: Dashboard — Change Detection

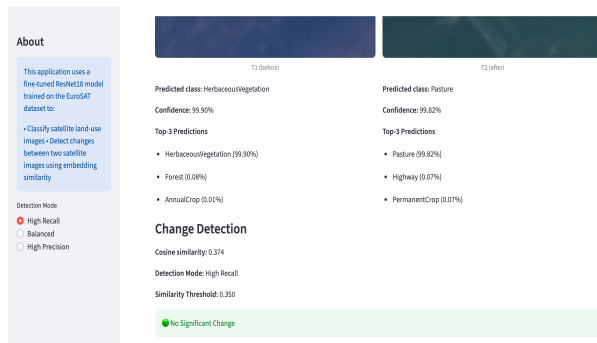
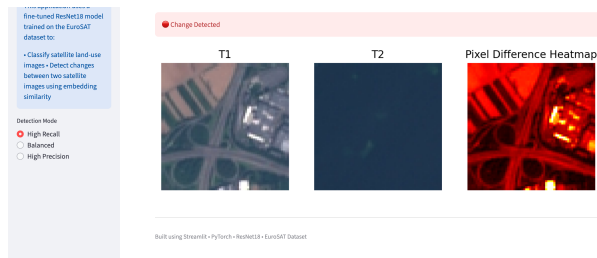


Figure 11: Dashboard — Change Heatmap



9. Conclusion & Limitations

Transfer learning substantially outperformed the scratch baseline (macro F1 0.948 vs 0.826), and unfreezing part of the backbone gave a further meaningful gain over head-only training. The model generalizes poorly to UC Merced under a strict exact-label-match evaluation, which is expected given the near-total mismatch between the two datasets' label vocabularies rather than a weakness in the fine-tuning itself. The simulated change-detection pipeline separates changed from unchanged pairs well (TPR 0.936 at FPR 0.062), though this is measured on synthetic pairs rather than real temporal imagery. The spatial leakage experiment did not confirm the expected direction of the leakage hypothesis in this single run, and would benefit from repetition across multiple seeds before drawing a firm conclusion. Overall, the project demonstrates a complete, working pipeline from data preparation through an interactive deployed dashboard, with its key simplifying assumptions (spatial block approximation, simulated T1/T2 pairs) clearly documented rather than hidden.